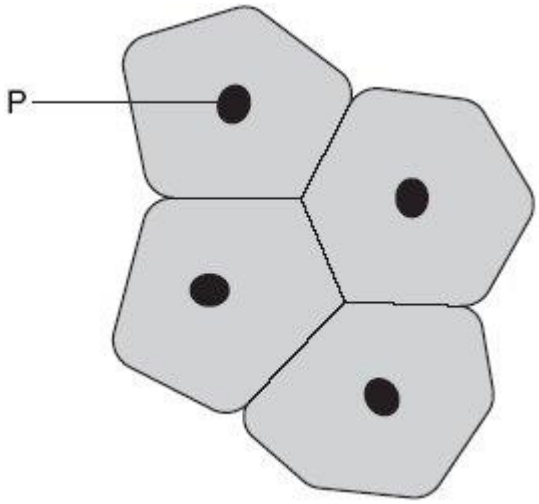
Q1.

(a) The diagram shows a group of cells from the lining of the mouth.



(i) Give the name and function of part P.

name of part P

.....

1 mark

function of part P

.....

.....

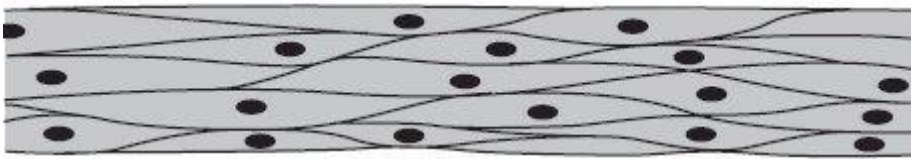
1 mark

(ii) Which word describes this group of cells?
Tick the correct box.

compound	organism	organ	tissue
☐	☐	☐	☐

1 mark

(b) The diagram below shows muscle cells from the wall of the human intestine.



(i) Muscle cells can contract.

Give **one** reason why muscles are needed in the intestine.

.....

.....

1 mark

(ii) Other cells in the intestine produce enzymes.

What effect do enzymes in the intestine have on nutrients such as protein?

.....

.....

1 mark

(iii) Which of the following is required in the diet to keep food moving through the intestine?
Tick the correct box.

fat	☐	fibre	☐
protein	☐	starch	☐

1 mark
maximum 6 marks

Q2.

(a) Egg white contains a protein called albumen.

A pupil carried out an experiment to investigate the digestion of three identical cubes of cooked egg white using the enzyme pepsin. She set up the experiment as follows.

Into beaker A, she put 50 cm^3 pepsin solution and a cube of egg white left whole.

Into beaker B, she put 50 cm^3 pepsin solution and a cube of egg white which had been cut into eight small cubes.

Into beaker C, she put 50 cm^3 pepsin solution and a cube of egg white which had been cut into sixty four tiny cubes.

She added 1 cm^3 hydrochloric acid to each beaker and placed the three beakers in a waterbath at 37°C. The table gives the pupil's observations five hours later.

beaker	observations five hours later
A	a little of the egg white had gone from the edges of the whole cube
B	the cubes were about half their original size
C	only a few tiny fragments of egg white were left

(i) Why was the cube of egg white which had been cut into sixty four pieces digested most quickly?

.....

.....

1 mark

(ii) When protein is digested it is broken down into smaller molecules. What name is given to these smaller molecules?

.....

1 mark

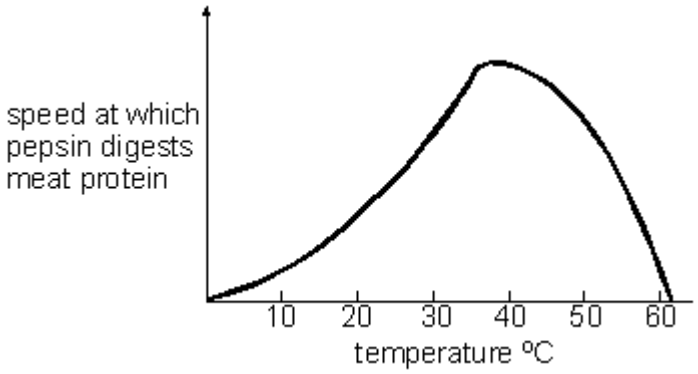
(b) In another experiment, it was found that pepsin does not digest sucrose (sugar). Explain why sucrose is not digested by pepsin.

.....

.....

1 mark

(c) The graph below shows how quickly pepsin digests meat protein at different temperatures.



Explain why pepsin does **not** digest protein at temperatures above 60°C.

.....

1 mark

(d) Glands in the stomach wall produce mucus, hydrochloric acid and an inactive form of pepsin called pepsinogen. In the stomach hydrochloric acid reacts with the pepsinogen to produce pepsin.

(i) Suggest **one** other purpose of acid in the stomach.

.....

.....

1 mark

(ii) Suggest **one** reason why pepsin is secreted in its inactive form.

.....

.....

1 mark

(iii) Suggest **one** purpose of mucus produced in the wall of the stomach.

.....

.....

1 mark

(iv) Bile from the liver makes the duodenum alkaline. Give **one** reason why the contents of the duodenum need to be alkaline.

.....

.....

1 mark

Maximum 8 marks

Q1.

(a) (i) • *name:* nucleus

1 (L6)

• *function:* it controls the cell

accept 'nuclei'; 'it contains genetic information'; 'contains chromosomes'

'it is the brain of the cell' is insufficient

1 (L6)

(ii) tissue ✓

if more than one box is ticked, award no mark

1 (L6)

(b) (i) any **one** from

• to push food **or** waste along

accept 'for peristalsis'

• to break up large pieces of food

accept 'mechanical digestion'

• to churn food

accept 'to mix in enzymes'

'to digest food' is insufficient

1 (L6)

(ii) • they speed up digestion

accept 'they digest them'

accept 'they break them down'

accept 'they turn it into amino acids'

1 (L6)

(iii) • fibre ✓

if more than one box is ticked, award no mark

1 (L5)

[6]

Q2.

(a) (i) larger surface area

*do **not** accept 'already partly broken down'*

1

(ii) any **one** from

• polypeptides

*do **not** accept 'monomers'*

• peptides

- amino acids
  *do **not** accept 'acids'*

1

(b) any **one** from

- enzymes are specific
  accept 'enzymes are selective'
- enzymes only act on one type of substrate
- pepsin only digests proteins
  accept 'it only digests protein'
  ***or** 'pepsin is a protease'*
- sucrose is not a protein
  *do **not** accept 'it is not a protein'*
- because of the shape of the molecules
  accept references to lock and key mechanism
  accept 'other enzymes digest the sucrose'
  *do **not** accept 'pepsin does not digest sugar'*
  or *'sucrose is a sugar'* **or** *'sucrose doesn't need digesting as it goes through the gut wall'*

1

(c) enzymes become denatured **or** destroyed **or** deactivated
*do **not** accept 'enzymes are killed'* ***or*** *'it is too hot to react'*
or *'by the time it reached 60° the meat was already digested'*

1

(d) (i) any **one** from

- kills bacteria **or** germs **or** micro-organisms
- provides the correct pH for the action of pepsin
  accept 'hydrolysis of protein'

1

(ii) pepsin could digest **or** damage the cells of the stomach
*accept 'if it were made in its active form it would damage the cells **or** stomach' accept 'stops the enzyme working before it is secreted'*

1

(iii) protects the stomach wall from the action of pepsin **or** acid
*accept 'protects the stomach' **or** 'lines the stomach'*
accept 'lubricates the path of the food'

1

(iv) any **one** from

- the enzymes in the duodenum work in a alkaline medium
  accept 'it stops the action of pepsin'
- to neutralise acids from the stomach

1

[8]